Group 3 - SCSD - SC2006

Software Engineering

Project Mission Statement

Job Hack SG helps students and early-career jobseekers discover *where demand is growing* in Singapore and *which roles match their skills today*. The app combines sector insights (growth, vacancies, in-demand skills) with a personal matching engine over live job postings. Users can search and filter jobs, see matches based on their current skill profile, and view missing-skill gaps alongside suggested courses to upskill. An applications page allows them to apply to that particular role by routing them to the correct website. The goal is a clear, testable, end-to-end prototype that demonstrates sector specific guidance along with practical role recommendations, built primarily on public datasets/APIs, specific to the Singapore context. What sets Job Hack SG apart from other competing platforms is that this will be a one stop portal for career related matters, and will cater specially for Singapore.

Functional Requirements

1. *The system shall allow users to create an account and add on their current skills to tune their job profile.*

    1.1. The system shall let users register and sign in with email/password.

    1.2. The system shall allow users, once signed - in, to edit profile fields: name, school, programme, graduation year.

    1.3. The system shall let users add, edit, delete skills (name; proficiency 1–5).

    1.4. The system also allows the user to do a Resume upload to ease the process of manually entering skills.

2. *The system shall be updated constantly with job related data.*

    2.1. The system shall ingest job postings daily (title, company, location in Singapore, description, tags/skills, posting date, apply URL, etc).

    2.2. The system shall compute and store a posting's active status if posted ≤ 60 days ago.

    2.3. The system shall ingest sector/skills summaries (e.g., vacancies by sector, top skills by sector) and store "sector → hot skills" pairs.

3. *The system shall display a Job Trends Dashboard.*

    3.1.The system shall display a Job Sector Trends dashboard: list sectors with indicators (e.g., vacancy index, growth direction) and their top skills.

    3.2. The system shall allow users to pick a sector and view top roles and top skills for that sector.

4. *The system shall have the ability of allowing users to search for jobs relevant to them.*

    4.1.The system shall allow users to do a keyword search over title/company/description and filters (location in Singapore, job type, recency).

    4.2. The system shall display a Matched Skills and Missing Skills chips for each recommended job.

5. *Course suggestions shall be made available to the users.*

5.1. For each job's top Missing Skills, the system shall display up to 3 relevant courses (title, provider, URL) for the user, that they can take up to improve their skillset.

6. *The system shall allow users to see details of jobs after searching for it.*

6.1. The system shall display a Job Details page with a full job description.

6.2. The system shall allow users to save/unsave a job.

6.3. The system shall allow users to apply for the posting by clicking on the link on the portal. It will route them to the company's website.

Non-Functional Requirements

1. *Usability*

1.1. The system shall be usable by both first-time and regular users.

1.2. The system shall allow a first-time user to complete onboarding (registration and add at least three skills) within 5 minutes without external help.

1.3. The system shall ensure all interactive UI components are fully keyboard - navigable.

2. *Performance*

2.1. The system shall load the home/dashboard page within 3 seconds under normal network conditions.

2.2. The system shall handle up to 500 concurrent users without degradation of service.

2.3. The system shall return search results in under 2 seconds for datasets containing ≤ 10,000 job postings.

2.4. The system shall generate and display personalized recommendations within 3 seconds for a user profile containing ≤ 40 skills.

3. *Security*

3.1. The system shall store user passwords using secure hashing and salting with an industry-standard algorithm.

3.2. The system shall automatically log out inactive users after 30 minutes.

3.3. The system shall virus scan all personally uploaded files before storage.

3.4. The system shall store uploaded files with restricted access permissions to prevent unauthorized usage by any third parties.

4. *Reliability and Availability*

4.1. The system shall have an uptime of at least 99% between 09:00 and 23:00 SGT daily excluding scheduled maintenance, which shall be announced beforehand.

4.2. The system shall recover from server failures within 5 minutes.

4.3. The system shall ensure no data loss in case of unexpected downtime by performing daily backups.

4.4. The system shall complete nightly ingestion jobs successfully on ≥ 95% of calendar days.

4.5. The system shall generate an admin alert in case of ingestion failure.

5. *Maintainability*

5.1. The system shall be designed in separate layers (front-end and back-end) so that changes in one do not break the other.

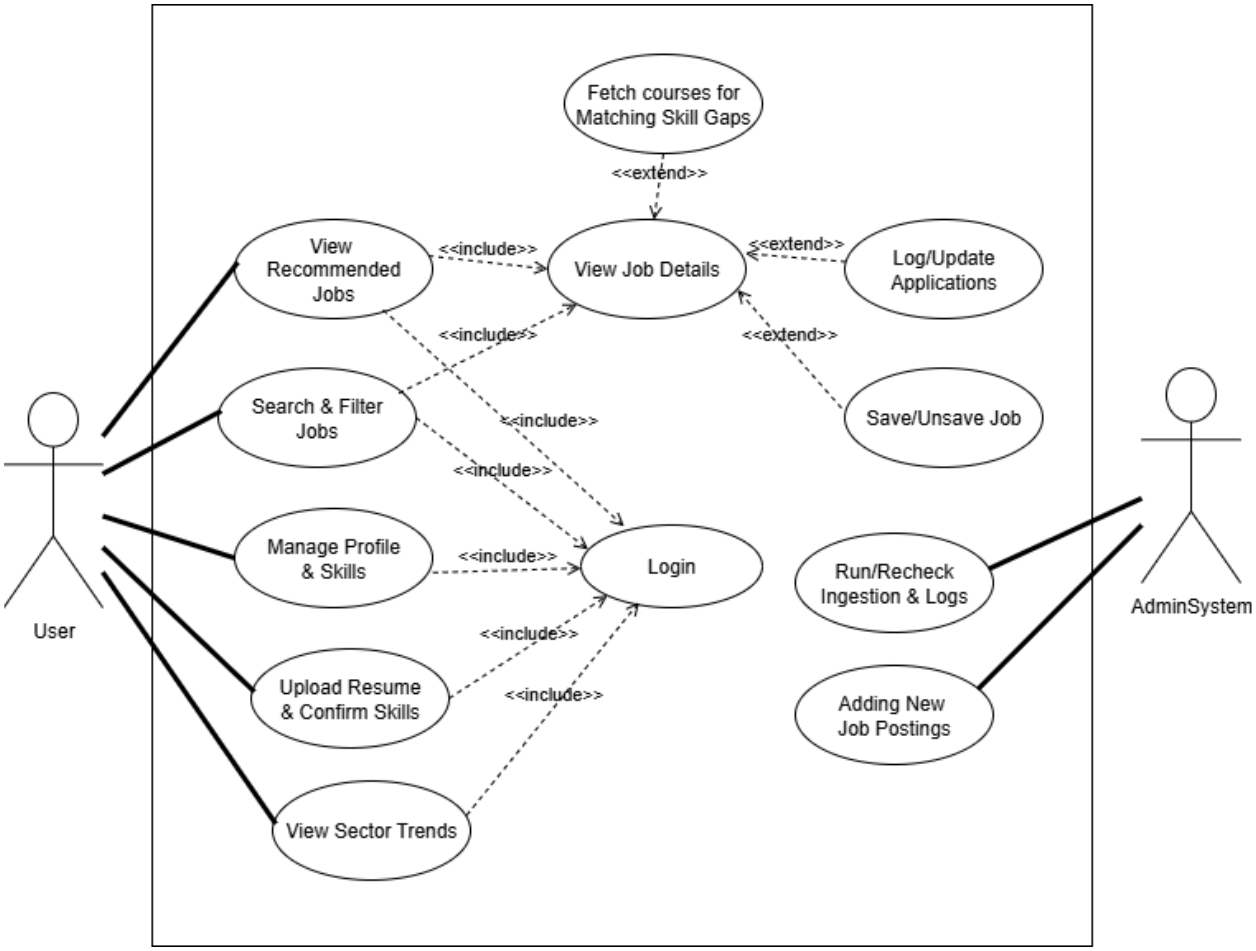
5.2. The core skill-matching logic shall achieve ≥ 60% unit test coverage by Lab 4.

5.3. The system shall allow updates to the data ingestion and matching components with less than 1 hour of downtime.

Data Dictionary

Name	Definition
User	A student or new jobseeker using JobHackSG. A user maintains a profile with personal details and skills and receives personalised information on sector trends and job recommendations.
Skill	A specific capability of a user (e.g., "Python Programming," "Data Analysis") that can be listed in their profile. It can be extracted from resumes, or identified as popular within a sector.
Resume	A digital document uploaded by the user that provides a description of their experience, educational qualifications, and skills. The system analyses the resume to pick out user skills and helps with profile building.
JobPosting	A structured representation of a job role sourced from live job boards or APIs. It includes the job title, employer, job description, location, required skills, and application link.
Sector	A labour market category in Singapore (e.g., ICT, Healthcare, Finance) with information such as growth trends, vacancies, and emerging skill demands.
SectorHotSkill	A skill classified as popular within a specific sector which can be identified from job posting analytics or public datasets. It is highlighted to users as a highly valued competency for better employability.
Recommendation	A personalised suggestion generated for a user, either a job or course, based on alignment of their skills with job/sector requirements. It includes skill gaps, and upskilling opportunities.
Course	An upskilling or training program (e.g., SkillsFuture, university modules) related to required skills in JobPostings. It is suggested to users to close identified skill gaps.
Application	A record of a user’s action to apply for a JobPosting. It is used to track when and where job applications are made and directs the user to the employer’s or application page.
IngestionRun	A logged process that retrieves, cleans, and updates job postings and sector data from external sources. It records metadata such as execution time, source, success/failure status, and error details.

Initial Use Case Model



Actors

- **User** (initiating): student/early-career jobseeker
- **AdminSystem**: operations
- **Jobs API** (external data source)
- **Courses API** (external data source)

Use Cases (UC)

1. **UC-1 Register & Login** *(included by others)*
2. **UC-2 Manage Profile & Skills**
3. **UC-3 Upload Resume & Confirm Skills**
4. **UC-4 View Sector Trends**
5. **UC-5 View Recommended Jobs**
6. **UC-6 Search & Filter Jobs**

7. **UC-7 View Job Details** *(included by UC-5/6)*
8. **UC-8 Save/Unsave Job** *(extends UC-7)*
9. **UC-9 Log/Update Application** *(extends UC-7)*
10. **UC-10 Fetch Courses for Matching Skill Gaps** *(extends UC-7)*
11. **UC-11 AdminSystem: Run/Recheck Ingestion & Logs**
12. **UC-12 AdminSystem: Adding New Job Postings**

Include/Extend:

- **UC-1** is **<<include>>** in UC-2/3/4/5/6.
- **UC-7** is **<<include>>** in UC-5/6.
- **UC-8 <<extend>>** UC-7; **UC-9 <<extend>>** UC-7; **UC-10 <<extend>>** UC-7.

Brief Use-Case Descriptions (will be refined further in Lab2)

Participating Actor: User (a jobseeker)

UC-1 Register & Login

Pre: For Login, User has an existing account. If there is no account, User is prompted to register instead
Post: User has logged into their account OR User has a newly created account and are logged in OR User is given a valid explanation for why their login failed

Main Flow:

1. System displays fields for email and password
2. User enters their details into the field.
3. User presses the login button.
4. System verifies their email and password with the account information in the system database.
5. If the account is verified, system displays a message that says 'Login is successful!'
6. User is brought to the home page / landing screen.

AF-S1: If User has no pre-existing account

1. System informs User they have to register for an account
2. User fills in details for Full Name, Email and Password.
3. User submits these details.
4. If it is valid, System takes these details and adds it to the system database.
5. System then displays a message to the User that the "Account has been created!"
6. Return to Step 1.

AF-S2: If the email or password is entered incorrectly

1. The system displays a message saying ‘Invalid email or password! Please Try Again!”
2. Return to Step 3.

UC-2 Manage Profile & Skills

Pre: Authenticated
Post: System displays the updated profile and skills.

Main flow:

1. User clicks on profile.
2. System displays any existing profile tied to the user, displaying the profile and skills
3. User clicks the option to manage profile and skills.
4. System displays the profile and skill fields that can be edited.
5. User enter their changes for the fields and confirms their changes.
6. System verifies these changes and confirms these changes to its database, displaying a message to the user “Your profile has been successfully changed!”

AF-S3: If any of User’s profile fields is empty

1. System displays an empty profile field, with a message that says “You have not added anything to this field! Add to it now!”
2. Return to Step 3.

UC-3 Upload Resume & Confirm Skills

Pre: Authenticated, existing resume to be uploaded
Post: System displays the updated resume and skills.

Main flow:

1. User clicks on profile.
2. System displays any existing profile tied to the user, displaying the profile and skills.
3. User clicks the option to update their resume.
4. System displays a message to “Upload your resume here!” and button to upload said resume.
5. System displays a preview of the resume and User confirms their changes if happy.
6. System verifies these changes and confirms these changes to the resume and skills to its database, displaying a message to the user “Your resume has been successfully updated!”

AF-S4: If User’s resume is empty

3. System displays an empty resume page, with a message that says “You have not added your resume! Add to it now!”
4. Return to Step 3.

UC-4 View Sector Trends

Pre: Authenticated; sector data exists in the database.
Post: System displays the data list of the selected sector.

Main Flow:

1. User opens “Trends” .
2. System lists sectors based on vacancy/growth and top skills.
3. User selects a sector.
4. System checks its database for the top roles and most suited skills for that sector.
5. System displays the top roles and most suited skills for that sector.

AF-S5: If there is no data in the selected sector

1. System displays a blank field with a message saying “No data found!”
2. Return to Step 2.

UC-5 View Recommended Jobs

Pre: Authenticated; existing job postings, user has at least 1 skill.
Post: System displays ranked jobs filtered by the skills.
Main Flow:

1. User selects “For You”.
2. System computes a match with matched/missing skills.
3. System displays a list of jobs recommended for user.
4. User scrolls through the list of jobs to select which one they like.
5. User selects a job.
6. See **UC-7**.

AF-S6: If no jobs are matched to the skills OR User has no skills

1. System displays a message “You have not added any skills!”
2. System displays generic trending jobs below the message.

UC-6 Search & Filter Jobs

Pre: Authenticated, existing job postings

Post: System displays results reflecting query/filters.
Main flow:

1. System displays list of unfiltered jobs based on date updated.
2. User opens the search / filter option box.
3. User selects and enters filters or searches based on key words
4. System searches database for jobs matching filters / key words
5. System displays to User list of jobs based on filters / key words.
6. User selects a job.
7. See **UC-7**.

AF-S7: If there is no available jobs based on the key word / filter

1. System displays a message saying “No jobs matched! Use another keyword or select less filters”
2. Return to Step 2.

UC-7 View Job Details

Pre: Authenticated; job active.
Post: System displays the details of the job; user may save/log application.
Main flow:

1. User clicks on job.
2. System checks the database for the job details.
3. System displays the job details.
4. User can apply for the job. See **UC-9**.

AF-S8: if user does not wish to apply but to view job later

1. User can save jobs as a bookmark to the Saved section.
2. See **UC-8**.
3. User clicks on the “Saved” button at the bottom.
4. System displays the saved jobs.

AF-S9: if job is removed

1. System displays a message saying “Job has been removed!”
2. Returns to **UC-5** or **UC-7** depending on the previous page User was on.

UC-8 Save/Unsave Job

Pre: Authenticated; on existing Job Detail page, job active
Post: System stores / removes the selected job in the Saved section
Main flow:

1. On the job's details page, User clicks on the bookmark button at the upper right corner.
2. System stores the selected job in the "Saved" section..
3. User goes to the saved jobs page.
4. System displays saved jobs to user.
5. User clicks on a saved job to go its detail page.
6. User again clicks on the bookmark button.
7. System removes the selected job in the "Saved" section.

AF-S10: if job has been removed

1. System displays a message saying "Job cannot be saved, it has been removed!" when user tries to save it.
2. Return to Step 1.

UC-9 Log/Update Application

Pre: Authenticated; on existing Job Detail page, job active
Post: Application submitted OR recorded in the "Apps" section

Main flow:

1. User clicks on the "Apply on company site" button.
2. External browser will be called to display the website.
3. User complete the application on the website.
4. User returns to the app and system updates the job with an 'Applied' tag.
5. User can record the application.
6. See **AF-S11**.

AF-S11: User records the application

1. User clicks on the "Log application" button.
2. System stores the application in the "Apps" section.
3. User clicks on the "Apps" button at the bottom.
4. System displays all saved applications.
5. User clicks on the "Edit notes" button to add the details / remarks of the application.

UC-10 Fetch Courses for Matching Skill Gaps

Pre: Authenticated; on existing Job Detail page, job active, existing skills in profile
Post: Display the courses to supplement skill gaps

Main flow:

1. User clicks on the option to view courses for skill gaps.
2. System checks the database for matching courses to supplement skill gaps.
3. System displays any results and corresponding links to the User.
4. User can choose to click on the links to the courses.

AF-S12: if no courses available for skill gaps

1. System displays a message saying that “There are no available courses!”
2. Return to the Job Details page.

UC-11 AdminSystem: Run/Recheck Ingestion & Logs

Pre: Admin Authenticated, Existing logs.
Post: Ingestion triggered or logs displayed.
Main flow:

1. Admin runs/rechecks ingestions and logs.
2. System runs through all the logs and checks for error
3. If no error is found, the system displays a message saying “Everything is fine!”

AF-S13: if there is 1 or more error:

1. The system displays a message, informing the admin of it and its details.
2. Admin makes a correction.
3. Repeat Step 1.

UC-12 AdminSystem: Adding New Job Postings

Pre: Admin Authenticated. Available job postings to be added.
Post: New job postings are added to the database.

Main Flow:

1. Admin checks the job postings database.
2. Admin adds the new job postings to the database.
3. System confirms jobs posting details and adds to the database.
4. System displays new job postings on all the lists to the user.

AF-S14: if new jobs postings have missing fields or error

1. System displays a message indicating “Error! New job posting has missing or invalid details!”

2. Admin fixes any mistakes on the job posting and adds it to the system again.
3. Return to step 3.

UI Mockups:

1. Onboarding:

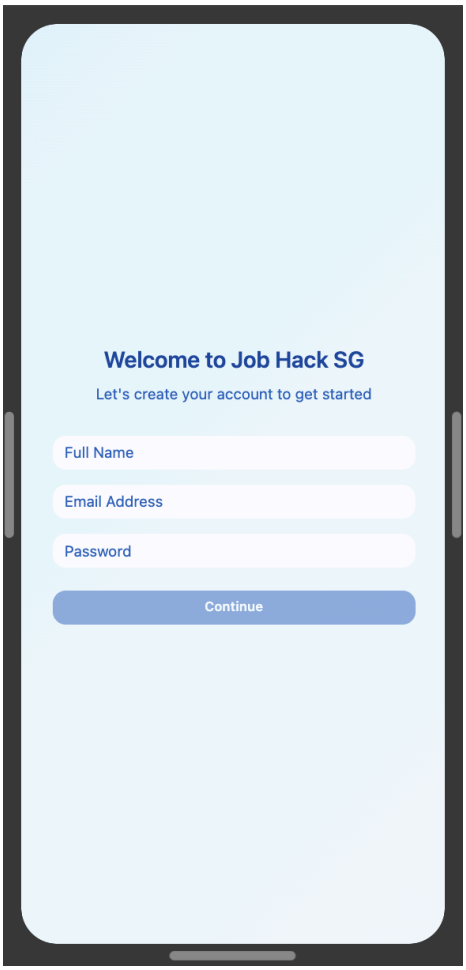


Figure 1.1

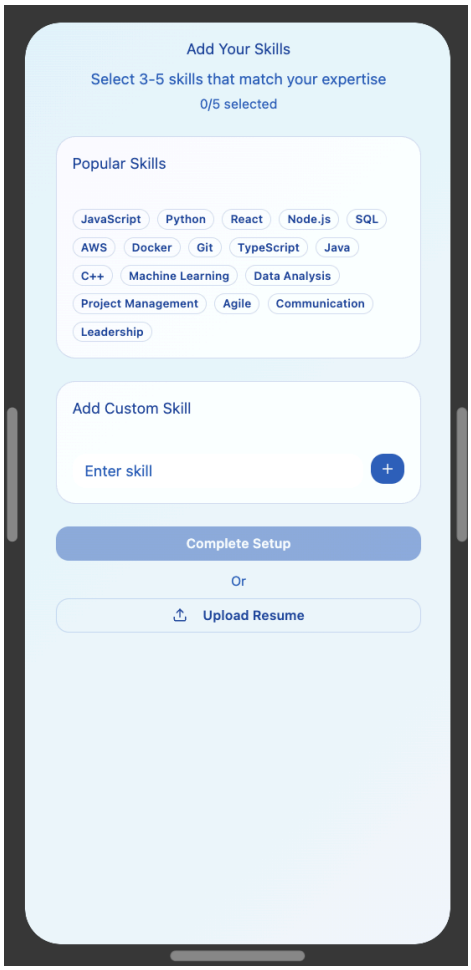


Figure 1.2

Figure 1.1 shows the Login Page of the application, where users are prompted to create a new account by providing their name, email address, and password. Figure 1.2 shows the Profile Creation Page, which enables users to select relevant skills from a predefined list of popular skills or add custom skills manually. Additionally, the application allows users to upload their resume to complete the account setup.

2. For You Page

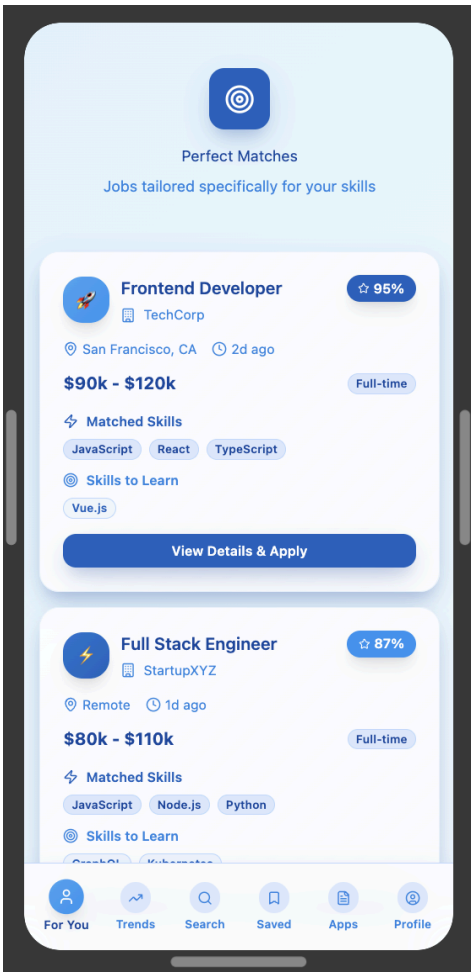


Figure 2.1

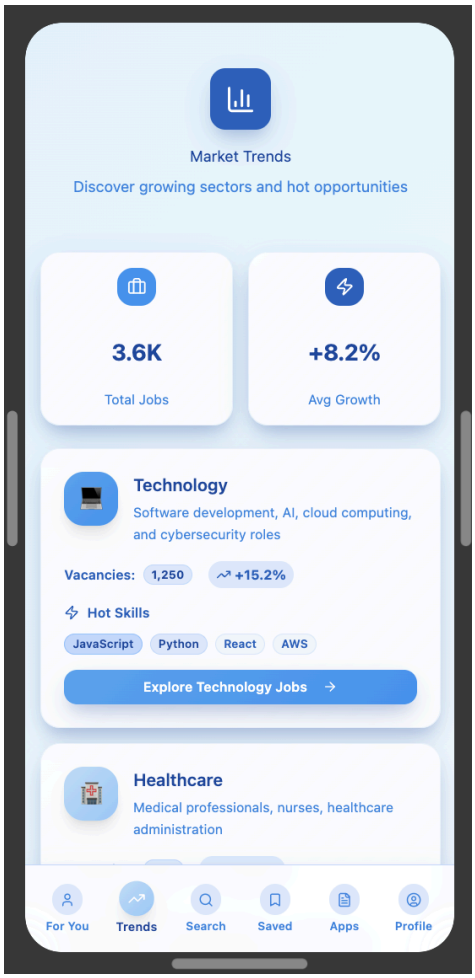


Figure 2.2

Figure 2.1 shows the Home Page, where job listings tailored to the user's profile are displayed. Figure 2.2 shows the Trends Page, where users can view job market trends categorized by different industries along with the popular skills each industry is looking for.

3. Search Page

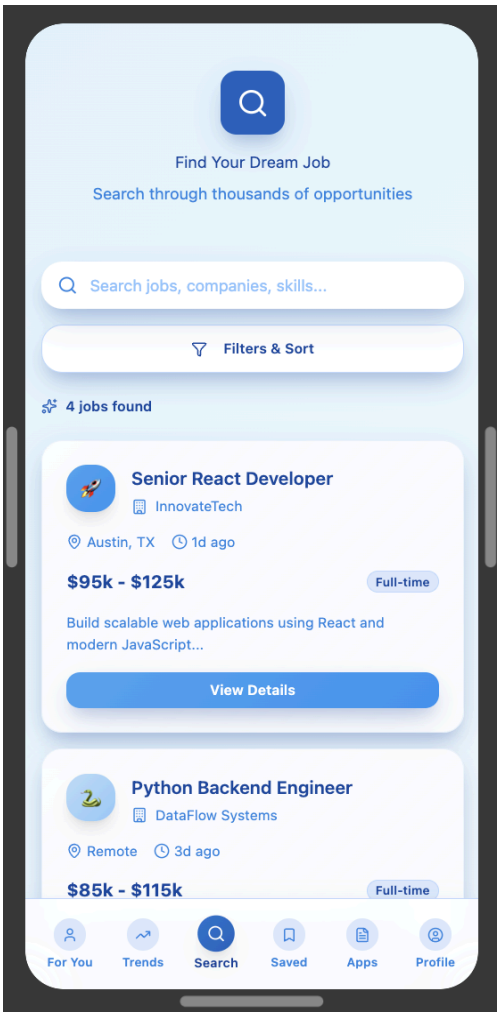


Figure 3.1

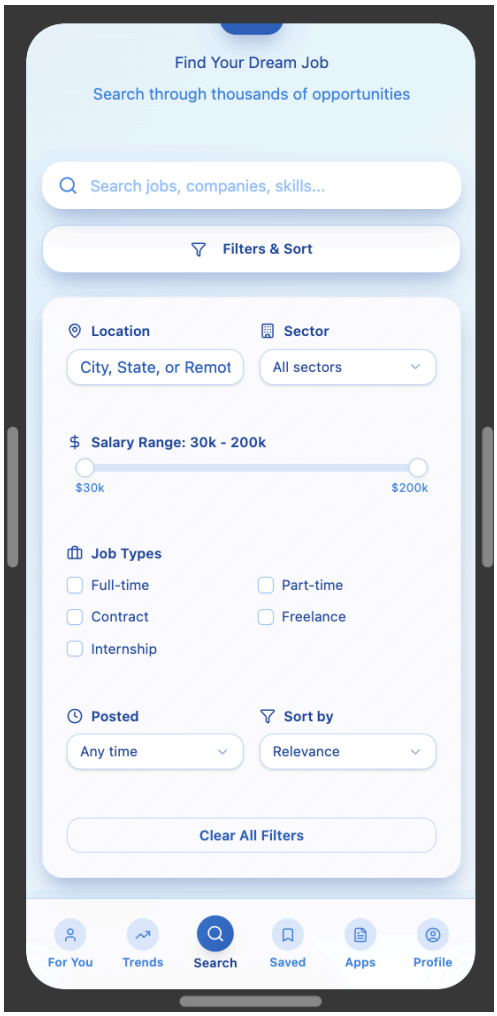


Figure 3.2

Figure 3.1 shows the job search page, which allows users to search for job opportunities that match their specific interests. Figure 3.2 shows the filter and sort functionality, enabling users to refine results by location, sector, salary range, and job type, as well as sort postings by the date they were advertised.

Note: While the prototype currently displays some sample jobs with overseas locations, the actual application is scoped to roles within Singapore only.

4. Job Details

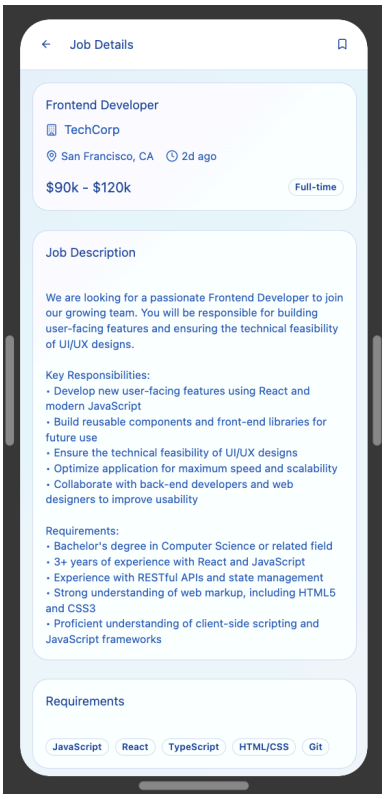


Figure 4.1

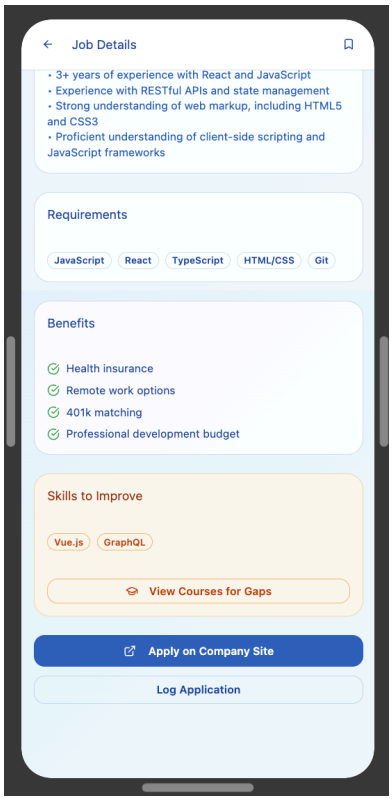


Figure 4.2

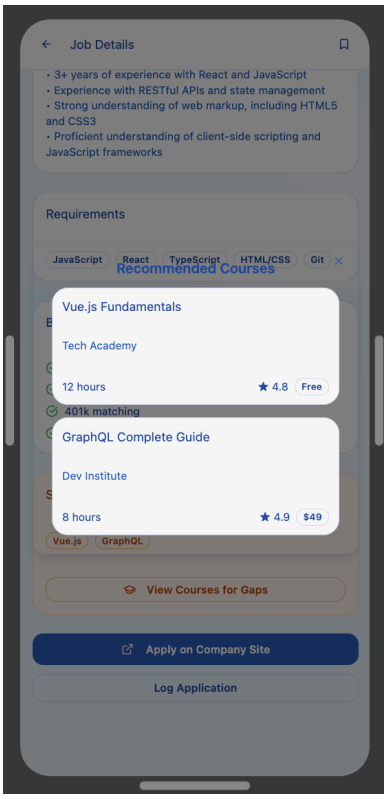


Figure 4.3

Fig 4.1 shows the Job Details page for a selected role. The page provides comprehensive information about the position, including the job title, company name, location, posting date, salary range and employment type. Users can read the full job description and key responsibilities of the job. Fig 4.2 details the required skills for the job, the benefits as well as highlighting “Skills to Improve” along with an option to view suggested courses to help bridge the gap. It also shows the interactive options provided by the application. Users can choose to save this job for further perusal by clicking on the top left “save” button on the page, apply directly through the company’s site, or log their application for that job so they have their own reference of it on this app using the buttons on the bottom of the page. Fig 4.3 shows samples of the suggested courses that users can undertake to bridge the gap between their skills and the company’s requirements, as well as other relevant information such as the academy offering the course, the rating, cost and time taken to complete the course.

5. Applications:

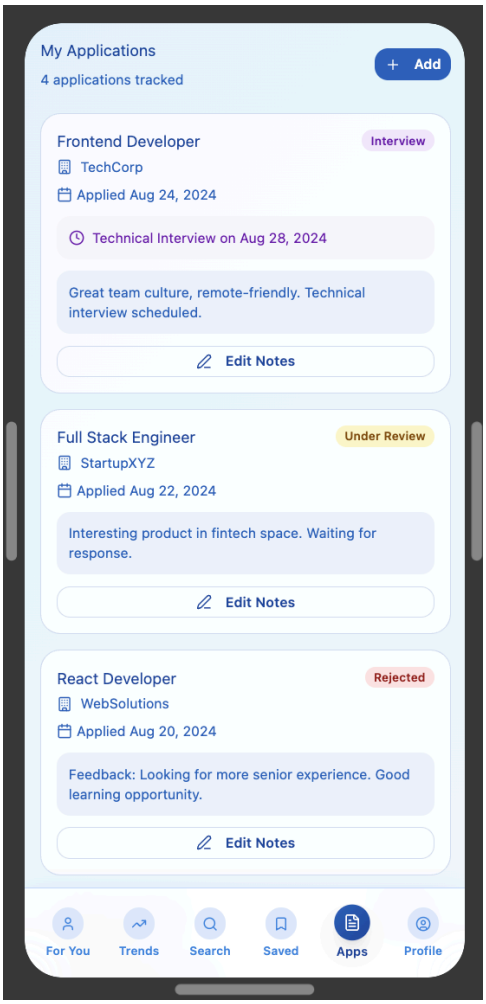


Figure 5.1

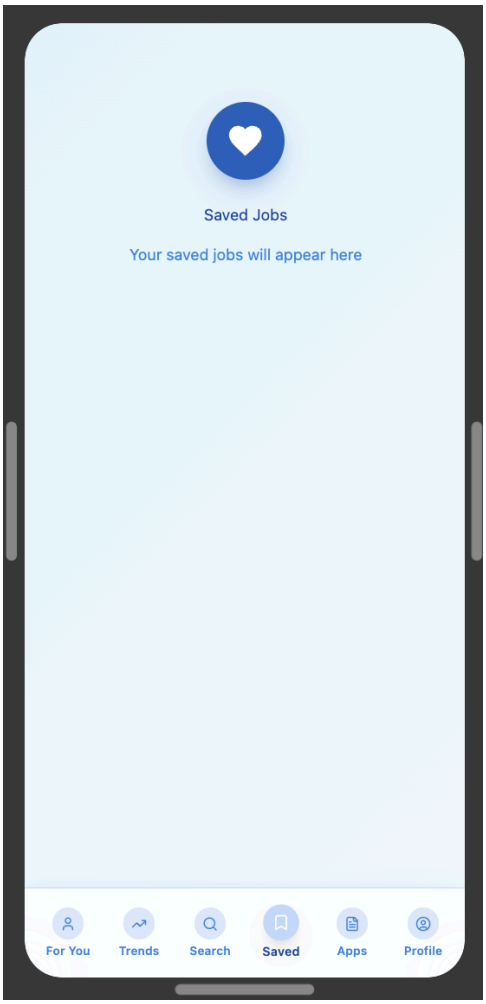


Figure 5.2

Fig 5.1 shows the My Applications page, where users can track all the roles they've applied for in one place. Each entry displays the job title, company name, application date, and current status - Applied, Under Review, Interview, Rejected. Additional details such as interview dates, recruiter feedback, or personal notes can be added and edited directly, allowing users to maintain a personalised record of their job search progress. Fig 5.2 shows the Saved page, which allows users to save jobs that they are interested in for future research.

Note: job applications are not submitted directly through this app; instead users apply directly via the respective company's site, and record the details here.

6. Profile:

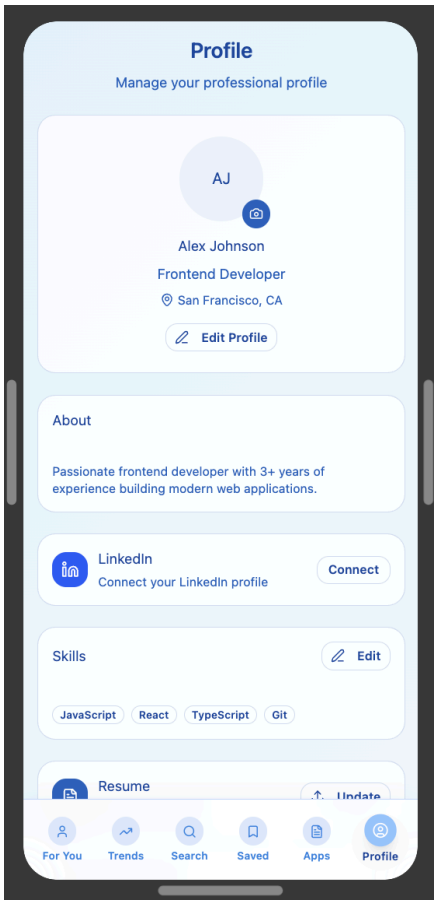


Figure 6.1

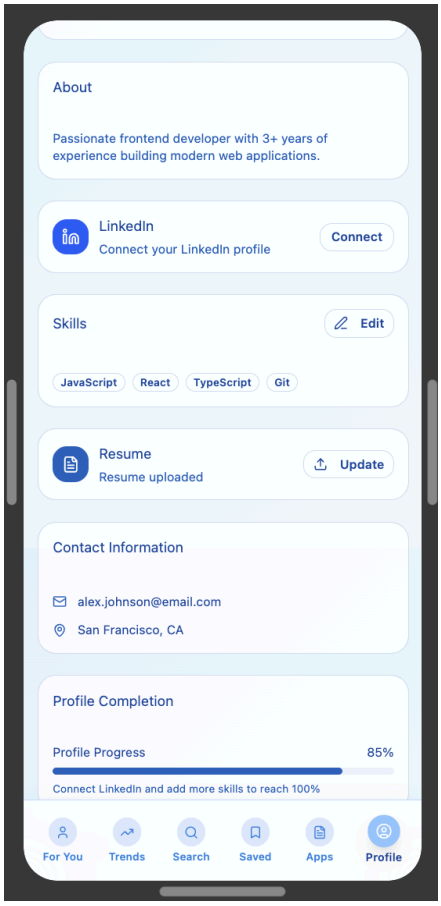


Figure 6.2

Fig 6.1 and 6.2 show the Profile page, where users manage their professional identity within the platform. Fig 6.1 showcases the user's basic details such as their name, role and location, as well as links to external profiles such as LinkedIn. It also highlights their list of Skills, with the ability to update it constantly if there are changes. Fig 6.2 has other key elements like resume upload and update functionality, and contact information for easy reference. A Profile Progress tracker motivates users to complete their profile by connecting LinkedIn or adding more skills.

Note: UC-11 is not included in the UI mockup because the mockups are intended to illustrate only the end-user's perspective, not the administrative or backend workflows. The functionality described in UC-11 is administrative in nature and falls outside the scope of the user-facing interface. Instead, UC-11 is documented in the system specification to ensure that the functionality is accounted for without cluttering the user-centric mockups.